VEHANT TECHNOLOGIES PRIVATE LIMITED

PSCDL Challenge 2026

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1. Challenge Overview

This document is the official specification for the **PSCDL 2026 Challenge**, organised by **Vehant Research Labs**. The challenge targets the design and implementation of an automated pipeline capable of identifying persistent changes in video footage of surveilled scenes. Participants will:

1. Develop an approach using publicly available datasets or their own curated data, and integrate it into the pipeline for detecting persistent changes.
2. Implement the required `generate_mask(p, c, video_path)` function.
3. Submit results evaluated against ground-truth masks.

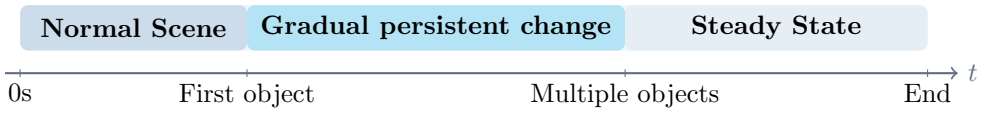
2. Dataset

Vehant Technologies provides a curated set of annotated videos, along with corresponding ground-truth masks and .txt files per video to illustrate expected pipeline behaviour.

3. Dataset Description

3.1. Video Structure

All videos in the dataset follow a consistent temporal pattern:



- **Phase 1 – Normal Scene:** In this duration no persistent object is added.
- **Phase 2 – Gradual Introduction:** Objects are introduced one by one at varying timestamps.
- **Phase 3 – Assimilation:** Persistent objects eventually get assimilated into the dynamic background after the cooldown period expires.

3.2. Mask Format

Ground-truth masks are **binary PNG images** with dimensions matching the source video frame:

- **White pixels** (`255`): Persistent change region.
- **Black pixels** (`0`): Background or no persistent change.
- A **blank mask** (all zeros) indicates no persistent change during that interval.

4. Pipeline Specification

Mandatory Interface

Participants **must** implement the following function exactly as specified.

```python
def generate_mask(p: int, c: int, video_path: str) -> None:
    """
    Generate per-second binary change masks starting from (persistent
        threshold + 10 seconds) for a given video, and save them in
       the `output_masks` directory.


    Parameters
    ----------
    p : int
        Persistence Threshold in seconds.  An object must remain
        continuously present for at least `p` seconds before it
        is classified as persistent change.

    c : int
        Cooldown Period in seconds.  Measured from the first
        introduction of an object.  Once `c` seconds have elapsed,
        the object is assimilated into the background and is no
        longer flagged.

    video_path : str
        Absolute or relative path to the input video file.

    Output
    ------
    Saves binary mask images to the 'output_masks/' directory
    (created automatically if it does not exist), one PNG per
    second of video:
        output_masks/mask_0001.png
        output_masks/mask_0002.png
        ...
        output_masks/mask_<N>.png
    """
    ...
```

Listing 1: Required function signature

5. Worked Example – Video 5

This section walks through the complete ground-truth derivation for **Video 5** to illustrate how parameters interact with the object timeline.

5.1. Video Metadata

Property	Value
Video Duration	417 seconds
Persistence Threshold (P)	60 seconds
Cooldown Period (C)	90 seconds

Table 1: Video 5 metadata

5.2. Scene Description – Object Introduction Timeline

#	**Object**	`Introduced` (t_0)	`Becomes Active` ($t_0 + P$)	`Cooldown` ($t_0 + C$)
1	Box	`93s`	`153s`	`183s`
2	Keyboard & Pan	`116s`	`176s`	`206s`
3	Telephone	`184s`	`244s`	`274s`
4	Towel / Cloth	`200s`	`260s`	`290s`
5	Thela (cart)	`373s`	`433s*`	`463s*`

*Exceeds video duration (417s); object never reaches persistence threshold.

Table 2: Object lifecycle timestamps for Video 5 with $P = 60$s, $C = 90$s

5.3. Ground-Truth Mask Derivation

Mask	Interval	Explanation
mask1.png	0s -- 152s	**Blank.** No object has yet met the persistence threshold. The box was introduced at 93s but needs until 153s to become active.
mask2.png	153s -- 175s	**Box only.** The box has been present for $\geq$ 60s and is now a persistent change. Keyboard & pan have not yet reached their threshold (introduced at 116s; active from 176s).
mask3.png	176s -- 182s	**Box + Keyboard + Pan.** Keyboard and pan become persistent. The box is still active (cooldown expires at 183s).
mask4.png	183s -- 205s	**Keyboard + Pan only.** The box reaches its cooldown at 183s and is assimilated into the background.
mask5.png	206s -- 243s	**Blank.** Keyboard and pan complete cooldown at 206s. Telephone (introduced 184s, active from 244s) is still transient.
mask6.png	244s -- 259s	**Telephone.** Telephone has persisted for $\geq$ 60s (from 184s) and becomes active at 244s.
mask7.png	260s -- 273s	**Telephone + Towel/Cloth.** Towel becomes active at 260s (introduced 200s + 60s). Both telephone and towel are persistent during this interval.
mask8.png	274s -- 289s	**Towel/Cloth only.** Telephone completes its cooldown at 274s and is assimilated into the background. Towel remains active until its cooldown expires at 290s.
mask9.png	290s -- 417s	**Blank.** Towel completes cooldown at 290s. Thela (introduced 373s) never reaches the persistence threshold before the video ends at 417s.

Table 3: Ground-truth masks for Video 5

5.4. Timeline Visualisation

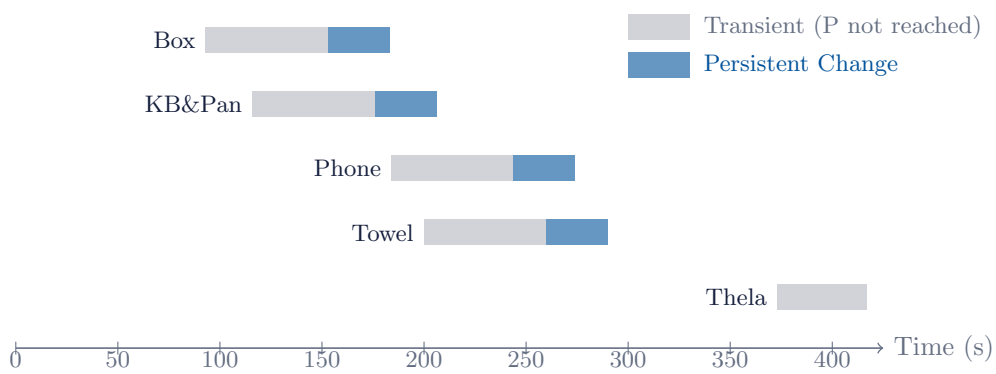


Figure 1: Per-object state timeline for Video 5. Grey = transient; blue = persistent change. Objects revert to background after cooldown (not shown separately).

6. Evaluation Methodology

6.1. Metric

Submissions are evaluated using the **F1 Score.**

Metric	Formula	Description
F1 Score	$\frac{2\lvert\hat{M} \cap M\rvert}{\lvert\hat{M}\rvert + \lvert M\rvert}$	Harmonic mean of precision and recall between the predicted mask $\hat{M}$ and the ground-truth mask M.

Table 4: Evaluation metric

6.2. Evaluation Protocol

The evaluation process will be conducted in multiple phases to ensure a fair and comprehensive assessment of the proposed approaches.

Phase 1 (Development): Participants are expected to utilize the provided sample dataset, along with the corresponding ground-truth annotations, to develop, test, and refine their pipeline.

Phase 2 (Final Testing): In the later stage of the challenge, a set of final test videos will be released without ground-truth masks. Participants are required to run their pipeline on these videos and generate the corresponding output masks. These outputs will constitute the final submission.

Blind Evaluation: In addition to the visible test sets, the submitted approach may also be evaluated on a separate blind test set that will not be shared with participants. This evaluation will be conducted internally to assess the generalization and robustness of the proposed solution.

7. Contact Information

For any query please contact: contest@vehant.com

Please use [**PSCDL 2026 Challenge**] as the subject line prefix in all correspondence.